AQA GCSE Physics
Topic – Atomic Structure
Section 1: detailed revision notes

1. The Structure of the Atom

Definition of an Atom

An **atom** is the smallest part of an element that can exist and still retain the properties of that element.

2. Subatomic Particles

Atoms are made of three main subatomic particles:

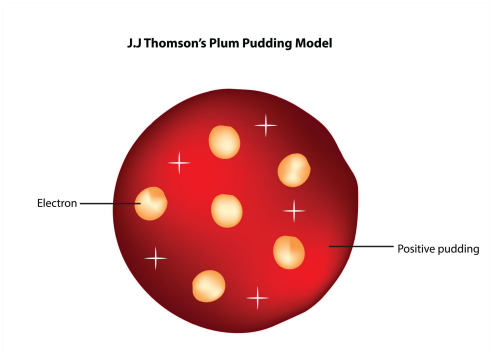
Particle	Charge	Relative Mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	–1	1/1836	Electron shells

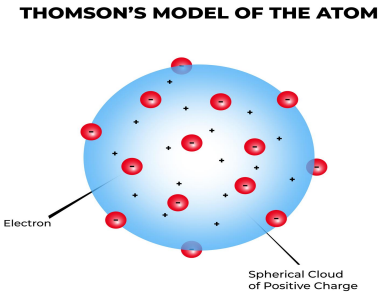
Key Exam Phrase

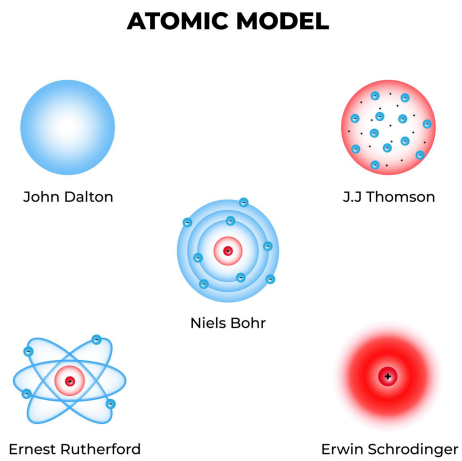
The nucleus is small, dense and positively charged, containing most of the atom's mass.

3. The Nuclear Model of the Atom

Early Model – Plum Pudding Model



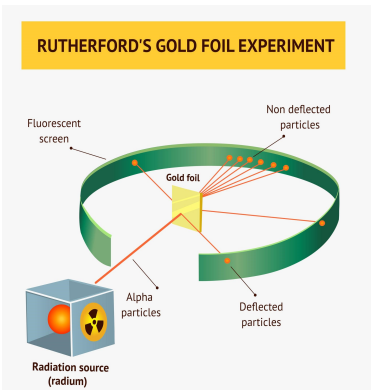


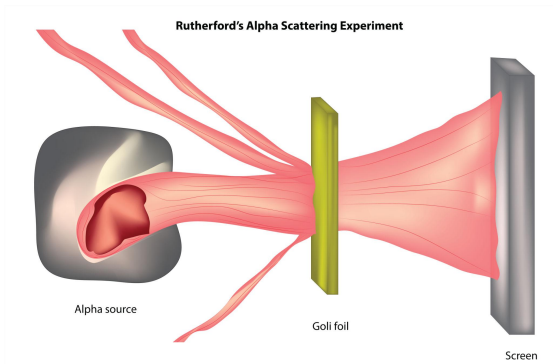


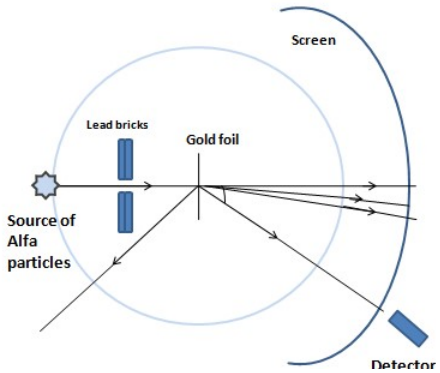
Proposed by **J. J. Thomson** in 1897.

- Atom was a ball of positive charge.
- Electrons were embedded within it.
- No nucleus.

Rutherford's Alpha Scattering Experiment







Conducted by **Ernest Rutherford**, with **Hans Geiger** and **Ernest Marsden**.

Observations:

- Most alpha particles passed straight through.
- Some were deflected slightly.
- A very small number bounced back.

Conclusions:

- Atoms are mostly empty space.
- A small, dense, positively charged nucleus exists.
- Most of the atom's mass is concentrated in the nucleus.

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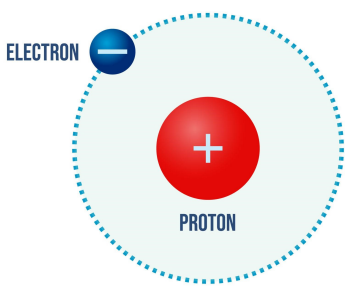
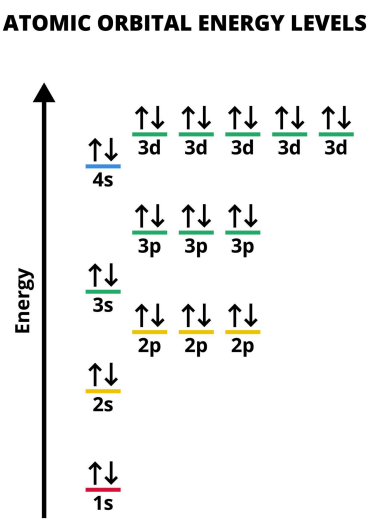
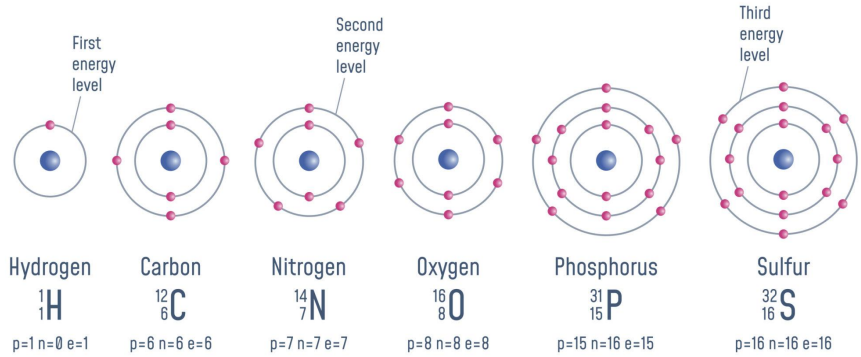
4. Bohr's Mode







Proposed by **Niels Bohr**.

- Electrons orbit the nucleus in fixed energy levels (shells).
- Electrons can move between shells by absorbing or emitting energy.

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5. Atomic Number and Mass Number

Atomic Number (Z)

The number of **protons** in an atom.

- Determines the element.
- In a neutral atom:

  Number of protons = Number of electrons

Mass Number (A)

The total number of protons and neutrons.

Equation:

[\textbf{Mass Number = Number of Protons + Number of Neutrons}]

[A = p + n]

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6. Isotopes

Definition

Atoms of the same element that have:

- The same number of protons
- Different numbers of neutrons

Example: Carbon Isotopes

- Carbon-12
- Carbon-14

Key Exam Phrase

Isotopes have the same chemical properties but different physical properties due to different masses.

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7. Electron Arrangement

Electrons fill shells in order:

- First shell → 2 electrons
- Second shell → 8 electrons
- Third shell → 8 electrons (for GCSE level)

Example: Sodium (Atomic number 11)

Electron arrangement:
2,8,1

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8. Ions

Definition

An ion is a charged particle formed when an atom gains or loses electrons.

- Losing electrons → Positive ion
- Gaining electrons → Negative ion

Example:

$Na \rightarrow Na^{+} + e^{-}$

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9. Required Practical (If Applicable in Specification)

There is no required practical directly on atomic structure, but understanding of the **alpha scattering experiment** is essential.

Students may be asked to:

- Describe the experiment.
- Explain the conclusions.
- Compare atomic models.

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10. Key Equations to Memorise

[A = p + n]

[\text{Number of neutrons} = \text{Mass number} - \text{Atomic number}]

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11. Common GCSE Exam Mistakes

- Saying electrons are in the nucleus.
- Forgetting that atomic number = number of protons.
- Confusing isotopes with ions.
- Not stating that atoms are mostly empty space.

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12. High-Level Exam Answer Structure (6-Mark Question Tip)

When describing the development of the atomic model:

1. Start with the plum pudding model.
2. Describe alpha scattering observations.
3. Explain why the model changed.
4. Mention the nucleus.
5. Refer to Bohr's energy levels.
6. Use scientific vocabulary: *evidence*, *experimental results*, *model*, *conclusion*.

End of Section 1